

UV/Vis SPECTROPHOTOMETRY: Game-Changer For Battery Tech

The exploration of ultraviolet/visible (UV/Vis) spectrophotometry within the realm of battery technology reveals its indispensable role in enhancing our understanding and management of battery health and efficiency



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In our journey through the world of battery technology, we have discovered that ultraviolet/visible (UV/Vis) spectrophotometry is a game-changer. You might be wondering, “What exactly is that?” Well, it is a technique where light is passed through clear, liquid samples, covering the ultraviolet to visible parts of the electromagnetic spectrum.

Why is this important? It is vital for monitoring the evolution of electrolyte solutions in batteries, which is key to assessing their health and efficiency. Have you ever pondered how we can tell a battery’s health just by looking at it? It all boils down to the spectral changes over time, which unveil the degradation of electrolytes, influenced by various chemical and electrochemical reactions

during the battery’s lifespan.

By focusing on key attributes like absorption spectrum, colour, transparency, and chemical composition, UV/Vis spectrophotometry offers invaluable insights into electrolyte deterioration and its impact on battery performance. Many professionals, particularly those in quality control and research and development sectors, use this technique to align electrolyte samples with established industry colour scales, especially in the yellow spectrum, such as the yellowness index or APHA.

The quality of the electrolyte is a crucial determinant in a battery’s functionality and longevity. This is because the electrolyte plays a pivotal role in charge transport. When the electrolyte quality is subpar, the efficiency of the battery

suffers. By comparing the electrolyte’s spectra at various stages of the battery’s charge and discharge cycles against a reference spectrum of a pristine electrolyte, we can effectively track its degradation. The rate of change in the UV/Vis spectra inversely correlates with the electrolyte’s lifespan, highlighting the significance of this technique in the development of longer-lasting batteries.

Electrolyte impurities and battery health

When we delve into the intricacies of lithium-ion



(Photo courtesy: METTLER TOLEDO)